Question	Answer	Marks
1	**Defining the problem**	
	L is the independent variable and $\Delta\theta$ or temperature change/increase is the dependent variable **or** vary L and measure $\Delta\theta$ or temperature change/increase	**1**
	keep t constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • oil in a beaker/container (on a bench) • coil fully submerged in oil • (bulb of) thermometer in the oil • at least three labels from thermometer, coil or resistance wire, oil, beaker/container, clamp/stand, bench Do not accept other heating sources.	**1**
	method to determine V – diagram of workable circuit including: • power supply connected to wire • voltmeter positioned to measure V across the coil	**1**
	measure the initial and final temperature and find the difference $\Delta\theta$	**1**
	method to determine t, e.g. use stopwatch/timer **and** method to determine L e.g. use a rule(r) to measure L / length of wire **or** e.g. using number of turns and measure the diameter of the coil with rule(r) / calipers	**1**

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td rowspan="4">1</td><td>Method of Analysis</td><td></td></tr>
<tr><td>plot a graph of $\Delta\theta$ against $\frac{1}{L}$ or equivalent, e.g. $\frac{1}{L}$ against $\Delta\theta$
Do not accept logarithms.</td><td>1</td></tr>
<tr><td><table><tr><td>for $\Delta\theta$ against $\frac{1}{L}$</td><td>for $\frac{1}{L}$ against $\Delta\theta$</td></tr><tr><td>$K = \frac{AtV^2}{m \times \text{gradient}}$</td><td>$K = \frac{AtV^2 \times \text{gradient}}{m}$</td></tr></table></td><td>1</td></tr>
<tr><td><table><tr><td>for $\Delta\theta$ against $\frac{1}{L}$</td><td>for $\frac{1}{L}$ against $\Delta\theta$</td></tr><tr><td>$Z = -mK \times y\text{-intercept}$
or
$Z = -\frac{AtV^2 \times y\text{-intercept}}{\text{gradient}}$</td><td>$Z = AtV^2 \times y\text{-intercept}$</td></tr></table></td><td>1</td></tr>
</table>

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 precaution linked to hot oil / beaker / wire, e.g. use of gloves to prevent burns from oil **or** precaution linked to spillage of oil, e.g. perform experiment in a tray	
	D2 keep A **and** m **and** V constant	
	D3 use a micrometer to measure the diameter (d) of the wire **and** $A = \frac{\pi d^2}{4}$	
	D4 repeat measurements of d along the wire and average	
	D5 method to reduce heat loss e.g. add insulation around the container / add a lid to the container	
	D6 method to keep V constant, e.g. adjust / change a variable resistor / power supply to keep V or voltmeter reading constant	
	D7 use a balance to determine the mass of the oil **and** mass of oil = mass of (beaker + oil) − mass of beaker **or** place beaker on balance and zero balance, then add oil and read balance	
	D8 stir the oil for uniform temperature **or** keep the initial temperature (of oil) constant	
	D9 repeat the experiment for the same value of L and average $\Delta\theta$ / average temperature change	
	D10 relationship valid if a straight line is produced (passing through $\left(-\frac{Z}{mK}\right)$) Do not accept line passing through the origin.	
	D11 method to determine L accurately, e.g. measure length of unwound coil	

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